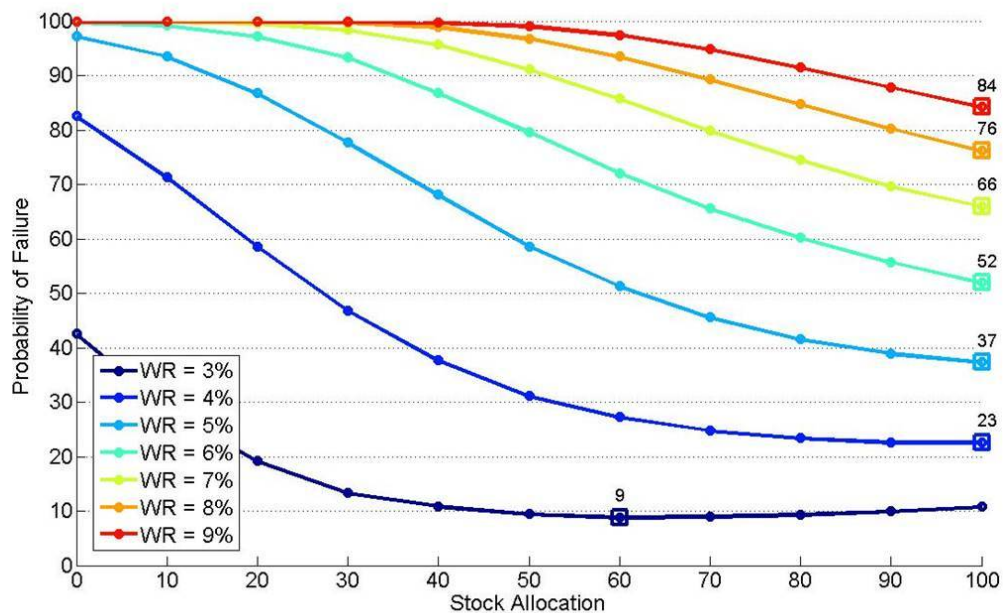


creates the only possible way for higher spending rates to work, but the odds for such success are not favorable; high spending rates are fundamentally unsustainable.

*Exhibit 5.7*

*Failure Probabilities for Inflation-Adjusted Withdrawal Strategies Over a 30-Year Retirement Horizon*

*Monte Carlo Simulations Starting from Current Market Conditions*



The previous discussion focused on low interest rates without accounting for any future impact from high stock market valuations. The risk premium that stocks earn above bonds was kept at its historical level. Exhibit 5.8 shows the type of risks created by the combination of high stock market valuations and low interest rates. In this case, we can observe the impact of both factors on sustainable retirement spending by looking to historical data rather than relying on Monte Carlo simulations.

An alternative way to view the historical data is to go beyond merely *considering* past withdrawal rate outcomes, by instead calculating how past outcomes relate to retirement date values of the underlying sources of returns. We can find a way to investigate these implications by following the approach described in 1998 by John Campbell and Robert Shiller, in their